Contents

1 Introduction

The Chagas assay is an Single Molecular Counting (SMC) immunoassay intended for qualitative detection of human Trypanosoma cruzi antibodies in plasma and serum specimens from human blood donors, including volunteer donors of whole blood and blood components, and other living donors. It is also intended in testing plasma and serum specimens to screen organ donors when specimens are obtained while the donor's heart is still beating, and in testing serum specimens to screen cadaveric (non-heart-beating) donors. It is not intended for use on cord blood specimens. The assay is not intended for use as an aid in diagnosis.

As stated in the Anti-T.cruzi Chagas Internal Performance Study (PGK-RD-PE-123456), the aim of this study was to evaluate the performance or agreement of the assay for a qualitative examination using an internal continuous response. This evaluation study was performed following the CLSI documents EP12 Ed3 “Evaluation of Qualitative, Binary Output Examination Performance”.

2 Procedure

The report has been automatically generated. Instruments and lots used for the examination are listed in Table 1. The instrument used in the study was qualified and verified according to PGK-PGK-GP-000041 (General Procedure for IQ/OQ/PQ of equipment) and PGK-PGK-GP-000001 (Control of Equipment of Measure, Inspection and Test). The quality controls of each run were verified. Data analysis has been executed by Conducted by.

Table 1. Description of instruments and lots

Instrument	Reagent lot	Expiration date	N
A	Lot 12	04/11/2024	9312
B	Lot 12	04/11/2024	5675
B	Lot13	04/11/2024	2691
B	Lot13	05/11/2024	8

For the evaluation of the agreement of Anti-T.cruzi assay, the PRISM assay has been used as the comparative method.

The ESA, ROCHE and Alinity assays have been used as supplemental testing to assess if the target condition (TC) was present or absent. Table 2 shows the number of specimens per assay.

Table 2. Supplemental test assays per study

Study	Supplemental test assay	Number of specimens
Specificity	Alinity	4
Specificity	ESA	6
Sensitivity	ESA	843
Sensitivity	ROCHE	92

The study was carried out following the Anti-T.cruzi assay Package Insert using 15797 specimens for specificity and 935 specimens for sensitivity.

In specificity analysis, repeatedly reactive specimens by Anti-T.cruzi assay were tested with a supplemental test. When the supplemental test result was reactive or indeterminate, the repeatedly reactive specimens were excluded from the specificity calculations; otherwise, repeatedly reactive specimens were considered as false positives.

In sensitivity analysis, all specimens were tested with both Anti-T.cruzi and a supplemental test. When the supplemental test result was non reactive or indeterminate, the specimens were excluded from sensitivity calculations. Non reactive specimens on Anti-T.cruzi assay that were reactive on supplemental test were considered as false negatives, and were further tested, for descriptive purposes only, with PRISM assay.

Each of 16732 total specimens were tested from February 13, 2025 to July 22, 2073. Different specimens were tested for sensitivity and specificity as described in Table 3 and Table 4, respectively.

Table 3. Description of specimens for specificity.

Sample matrix	Types of specimens	Collection center	N
Plasma	Archived	Access	8971
Serum	Archived	Access	6826

Table 4. Description of specimens for sensitivity.

Specimen category	Sample matrix	Types of specimens	Collection center	N
Endemic Population	Plasma	Archived	Access	546
Endemic Population	Serum	Archived	Access	51
Endemic Population	Serum	Fresh	Access	8
Endemic Population	Serum	Fresh	Barcelona	10
Parasite Positive	Plasma	Archived	Access	92
Parasite Positive	Plasma	Fresh	Access	20
Serology Positive	Plasma	Archived	Access	165
Serology Positive	Serum	Archived	Access	43

2.1 Acceptance Criteria

The internal performance will be considered acceptable if it meets the specification shown in Table 5.

Table 5. Acceptance criteria

Parameter	Criteria
Specificity[1]	$\geq$ 99.5%
Sensitivity[1]	$\geq$ 99.5%

[1]European Commission. (2022).

3 Experimental Data

3.1 Sample Size Calculation

The common specifications of guide [1]European Comission (2022) for Chagas assay are at least 5000 samples for specificity and 400 samples for sensitivity (Table 1. First-line assays: anti-T. cruzi of ANNEX XII).

CLSI EP12 Ed3 recommends the use of the Wilson score method for calculating the confidence intervals for sensitivity and specificity.

The sample size needed to achieve a 95% confidence interval for specificity of width 0.0202 with a confidence level of 0.05 for an estimated probability of success of 0.995 is 300.

The sample size needed to achieve a 95% confidence interval for sensitivity of width 0.0004 with a confidence level of 0.05 for an estimated probability of success of 1 is 10000.

The final sample for sensitivity is 935 and for specificity is 15797.

4 Analysis and Results

4.1 Specificity

Specimens were collected from donors (including first time donor) from 1 sites: Access. A total of 15797 specimens were analyzed, of which 8971 were plasma specimens and 6826 were serum specimens.

The specificity agreement between the Anti-T.cruzi assay and the PRISM was 99.92% (15785/15797).

According to the Anti-T.cruzi assay package insert, the 10 initially reactive (IR) specimens were retested in duplicate. After retesting in Anti-T.cruzi assay, 10 specimens were repetitive reactive (RR, at least one repetition was positive).

Supplemental testing (see Table 2) was performed on 10 specimens out of 10 RR specimens with Anti-T.cruzi assay. Table 6 shows the details on supplemental testing and the final decision for each specimen.

The specificity in blood donors was calculated to be 99.98% (15787/15790) with a 95% confidence interval of 99.94% to 99.99% (Table 7).

Table 6. Description of the specimens from donors (specificity) that were RR by Anti-T.cruzi assay.

Matrix	Sample ID	Chagas assay: Result (S/CO) (test1/test2/test3)	Chagas assay: RR/NR	Suppl test results R/NR/IND	Final Decision
Plasma	plasma1	1.50/1.50/1.50	RR	R	Remove
Plasma	plasma2	1.50/1.50/1.50	RR	IND	Remove
Plasma	plasma3	1.50/1.50/1.50	RR	IND	Remove
Plasma	plasma4	1.50/1.50/1.50	RR	NR	False positive
Serum	serum1	1.50/3.00/1.40	RR	NR	False positive
Serum	serum2	1.50/2.00/3.00	RR	NR	False positive
Serum	serum3	1.50/4.00/2.00	RR	R	Remove
Serum	serum4	1.50/8.00/5.00	RR	R	Remove
Serum	serum5	1.50/1.80/2.00	RR	IND	Remove
Serum	serum6	1.50/2.00/1.30	RR	IND	Remove

Abbreviations: IR = Initially Reactive; RR = Repeatedly Reactive; R = Reactive; IND = Indeterminate; S/CO = Signal to Cutoff; Suppl = Supplemental.

Table 7. Anti-T.cruzi assay specificity.

Matrix	Number tested	NR (% of Total) (95%CI)	IR (% of Total) (95%CI)	RR (% of Total) (95%CI)	RR samples without suppl test (% of RR)	Suppl test results among RR: Number R (% of RR)	Suppl test results among RR: Number NR (% of RR)	Suppl test results among RR: Number IND (% of RR)	Specificity (%) (95%CI)
Plasma	8971	8967 (99.96) (99.89 - 99.98)	4 (0.04) (0.02 - 0.11)	4 (0.04) (0.02 - 0.11)	0 (0.00)	1 (25.00)	1 (25.00)	2 (50.00)	99.99 (8967/8968) (99.94 - 100.00)
Serum	6826	6820 (99.91) (99.81 - 99.96)	6 (0.09) (0.04 - 0.19)	6 (0.09) (0.04 - 0.19)	0 (0.00)	2 (33.33)	2 (33.33)	2 (33.33)	99.97 (6820/6822) (99.89 - 99.99)
Total Donors	**15797**	**15787 (99.94) (99.88 - 99.97)**	**10 (0.06) (0.03 - 0.12)**	**10 (0.06) (0.03 - 0.12)**	**0 (0.00)**	**3 (30.00)**	**3 (30.00)**	**4 (40.00)**	**99.98 (15787/15790) (99.94 - 99.99)**

Abbreviations: IR = Initially Reactive; RR = Repeatedly Reactive; R = Reactive; NR = Non Reactive; IND = Indeterminate; CI = Confidence Interval; Suppl = Supplemental.

4.2 Sensitivity

Assay sensitivity was calculated on a total of 935 specimens, of which 823 were plasma specimens and 112 were serum specimens. Specimens were collected from 2 sites: Access and Barcelona. The specimens used to assess assay sensitivity were: preselected parasite positive (112), preselected serology positive (208) and individuals from endemic areas (615).

According to the Anti-T.cruzi assay package insert, the 467 initially reactive (IR) specimens were retested in duplicate. After retesting in Anti-T.cruzi assay, 467 specimens were repetitive reactive (RR, at least one repetition was positive).

Supplemental testing was performed on all specimens to assess target condition present or absent (see Table 2).

Additionally, just for reference, testing with PRISM assay was performed on 2 specimens out of 2 specimens that were reactive by supplemental test and non reactive by Anti-T.cruzi assay.

Table 8 shows the details on discrepancy specimens and the final decision.

The sensitivity in preselected positive specimens was calculated to be 100.00% (320/320) with a 95% confidence interval of 98.81% to 100.00% (Table 9).

The sensitivity of the Anti-T.cruzi assay in specimens from individuals from endemic areas was estimated to be 98.65% (146/148) with a 95% confidence interval of 95.21% to 99.63% (Table 9).

Table 8. Description of the specimens for sensitivity analysis that were RR by PRISM assay and NR by Anti-T.cruzi assay and specimens from endemic population that were NR by PRISM assay and RR by Anti-T.cruzi assay.

		Chagas assay				
Category	**Sample ID**	**Result (S/CO) (test1/test2/test3)**	**RR/NR**	**Suppl test results R/NR/IND**	**Comparator results with PRISM R/NR/IND**	**Final Decision**
Endemic Population	sample1	0.30/NA/NA	NR	R	R	False negative
Endemic Population	sample2	0.30/NA/NA	NR	R	NR	False negative
Endemic Population	sample469	1.50/1.50/1.50	RR	NR	Not provided	False positive (not used in sensitivity calculations)

Abbreviations: IR = Initially Reactive; RR = Repeatedly Reactive; R = Reactive; NR = Non Reactive; IND = Indeterminate; S/CO = Signal to Cutoff; Suppl = Supplemental.

Table 9. Anti-T.cruzi assay sensitivity.

		Suppl test results (number)			Anti-T.cruzi results						Comparator results with PRISM	
Category	**Number tested**	**R**	**NR**	**IND**	**NR (% of Total)**	**IR (% of Total)**	**RR (% of Total)**	**Among R by suppl test**	**Among NR by suppl test**	**Among IND by suppl test**	**On NR by Chagas and R by suppl test**	**Sensitivity (%) (95%CI)**
Serology Positive	208	208	0	0	0 (0.00)	208 (100.00)	208 (100.00)	208 RR + 0 NR	0 RR + 0 NR	0 RR + 0 NR	0 R + 0 NR + 0 IND + 0 No Data	100.00 (208/208) (98.19 - 100.00)
Parasite Positive	112	112	0	0	0 (0.00)	112 (100.00)	112 (100.00)	112 RR + 0 NR	0 RR + 0 NR	0 RR + 0 NR	0 R + 0 NR + 0 IND + 0 No Data	100.00 (112/112) (96.68 - 100.00)
Subtotal	**320**	**320**	**0**	**0**	**0 (0.00)**	**320 (100.00)**	**320 (100.00)**	**320 RR + 0 NR**	**0 RR + 0 NR**	**0 RR + 0 NR**	**0 R + 0 NR + 0 IND + 0 No Data**	**100.00 (320/320) (98.81 - 100.00)**
Endemic Population	615	148	467	0	468 (76.10)	147 (23.90)	147 (23.90)	146 RR + 2 NR	1 RR + 466 NR	0 RR + 0 NR	1 R + 1 NR + 0 IND + 0 No Data	98.65 (146/148) (95.21 - 99.63)
Total	**935**	**468**	**467**	**0**	**468 (50.05)**	**467 (49.95)**	**467 (49.95)**	**466 RR + 2 NR**	**1 RR + 466 NR**	**0 RR + 0 NR**	**1 R + 1 NR + 0 IND + 0 No Data**	**99.57 (466/468) (98.46 - 99.88)**

Abbreviations: IR = Initially Reactive; RR = Repeatedly Reactive; R = Reactive; NR = Non Reactive; IND = Indeterminate; CI = Confidence Interval; Suppl = Supplemental.

5 Conclusions

The specificity was 99.98% (99.94% to 99.99%) and the sensitivity was 99.57% (98.46% to 99.88%) (Table 10).

Table 10. Acceptance criteria evaluation

Parameter	Result	Criteria	Evaluation
Specificity	99.98%	$\geq 99.5\%$	PASS
Sensitivity	99.57%	$\geq 99.5\%$	PASS

The specificity and the sensitivity of the Chagas assay meets the acceptance criteria.

The specificity agreement between the Anti-T.cruzi assay and the PRISM was 99.92% (15785/15797).

6 Software

This report has been generated using the following software:

- SMC version 1.0.0
- R version 4.4.2 (2024-10-31 ucrt)
- tidyverse package version 2.0.0
- knitr package version 1.49
- kableExtra package version 1.4.0

7 Data Input Authors and Reviewers

Data input authors:

- Author3
- Author4
- Author1
- Author2

Data input reviewers:

- Reviewer3
- Reviewer4
- Reviewer1
- Reviewer2

8 References

CLSI EP12 Ed3 “Evaluation of Qualitative, Binary Output Examination Performance”

European Commission. (2022). Commission Implementing Regulation (EU) 2022/1107 of 4 July 2022 on performance studies for devices under Regulation (EU) 2017/746 of the European Parliament and of the Council on in vitro diagnostic medical devices. Official Journal of the European Union, L 173, 49–72.

PGK-RD-PE-123456 Performance Study Protocol